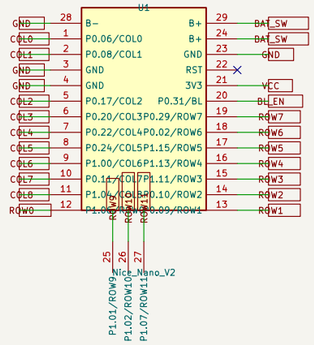


U1: user-installed TENSTAR / compatible nRF52840 module.
Module USB-C faces the rear case opening directly; no extension cable.
21 GPIO: 20 matrix + BL_EN. P0.09/P0.10 configured as GPIO.
No tall sockets; mechanical model defines mounting envelope.
Battery/charger and exact clone remain subject to bench validation.

Matrix

File: matrix.kicad_sch

Power

File: power.kicad_sch

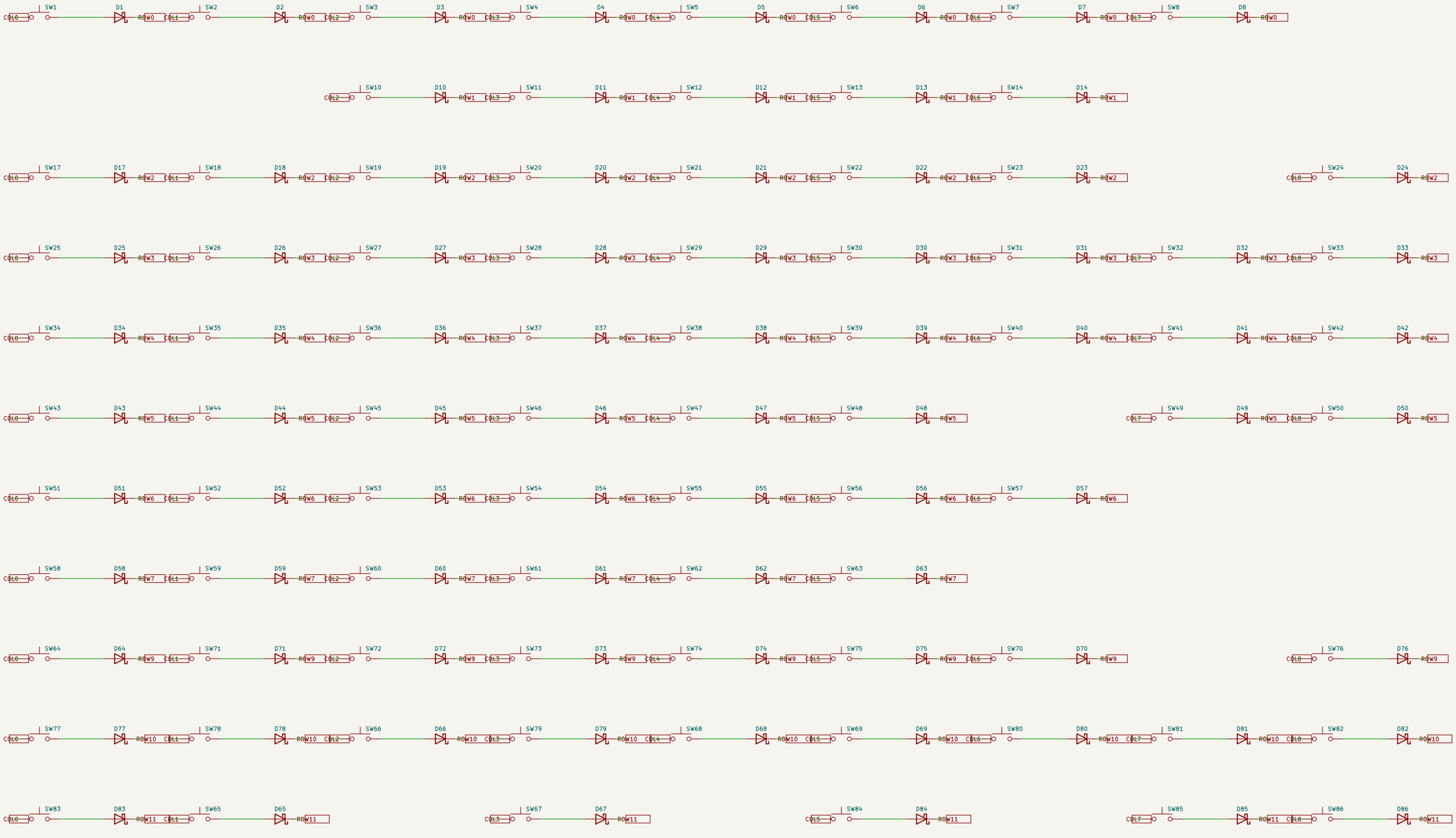
Pro Micro nRF52840 carrier; two layers; bottom assembly; module by user

Sheet: /
File: keyboard.kicad_sch

Title: TKL v9 — Controller

Size: A3	Date:	**Rev: 9**
KiCad E.D.A. 10.0.6		Id: 1/3

83 switches; SW9/SW15/SW16 removed; ROW8 unused. Remaining diodes: BAT46W retained; module GPIO is 3.3 V.

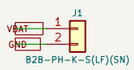


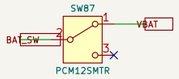
Pro Micro nRF52840 carrier; two layers; bottom assembly; module by user

Sheet: /Matrix/
File: matrix.kicad_sch

Title: TKL v9 – Matrix

Size: A3	Date:	**Rev: 9**
KiCad E.D.A. 10.0.6		Id: 2/3



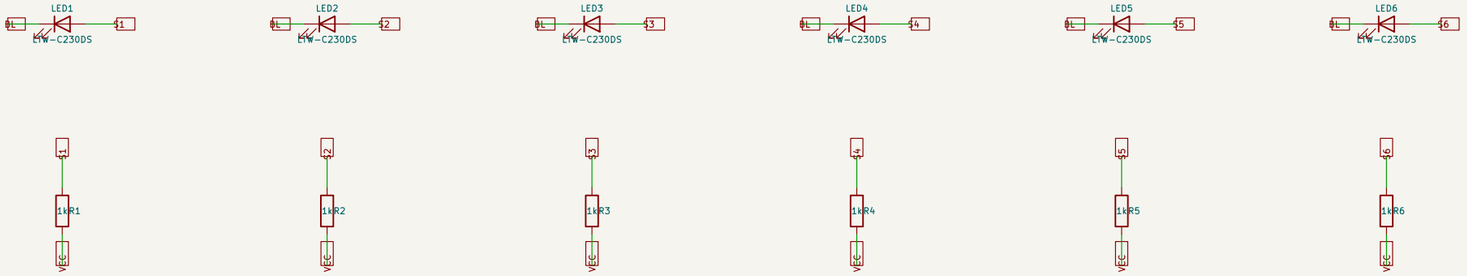


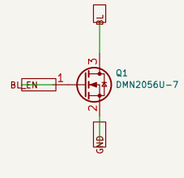


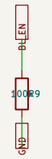












J1: protected 1S / 4.2 V pack, polarity must match.
SW87 disconnects battery from module. OFF also disconnects charging.
USB may still power the keyboard with SW87 OFF.
Only the module charger is connected; no separate BQ24072/NTC here.
Validate charge current, cell temperature protection and pack compatibility.
VCC is the module switched 3.3 V rail. Six 1k LED resistors: conservative initial brightness/current; Q1 provides common active-high PWM.

Pro Micro nRF52840 carrier; two layers; bottom assembly; module by user

Sheet: /Power/
File: power.kicad_sch

Title: TKL v9 – Power

Size: A3	Date:	Rev: 9
KiCad E.D.A. 10.0.6		Id: 3/3